

FINAL PROJECT REPORT

IBM Telco Customer Churn Analysis

Data Cleaning • EDA • SQL • Visualization • Statistics • Customer Segmentation
• Churn Prediction

Dataset: 7,043 customer records • 21 columns

Prepared as the final reporting deliverable for the Data Analytics Internship.

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This report consolidates the analytical work completed across the internship tasks. Numerical results are calculated from the cleaned Telco Customer Churn dataset used in the project.

1. Executive Summary

The project analyzes customer churn using the IBM Telco Customer Churn dataset containing 7,043 customer records and 21 variables. The overall churn rate is **26.5%**, meaning approximately one in four customers in the dataset discontinued the service.

The workflow combined data cleaning, exploratory data analysis, SQL-based extraction, Python visualization, statistical testing, customer segmentation with K-Means, and basic predictive modeling using Logistic Regression and a Decision Tree.

Statistical analysis found a strong difference in monthly charges between churned and retained customers (Welch's t-test $p < 0.001$). Contract type was also strongly associated with churn (chi-square $p < 0.001$). The average monthly charge was 64.76, with a 95% confidence interval of 64.06 to 65.46.

For customer segmentation, the strongest silhouette score among $K=2-10$ was obtained at **$K=6$** (silhouette score 0.471). For churn prediction, Logistic Regression achieved 77.9% accuracy and 0.816 ROC-AUC on the held-out test set, while the Decision Tree achieved 77.8% accuracy and 0.807 ROC-AUC.

Main business implication: retention efforts should prioritize customers on flexible month-to-month contracts, customers with higher monthly charges, and newer customers who have not yet developed long tenure.

2. Problem Statement & Objectives

Problem statement: Customer churn creates revenue uncertainty and increases the cost of acquiring replacement customers. The objective of this project is to analyze customer behavior, identify churn-related patterns, segment the customer base, and build a basic churn prediction workflow that can support retention decisions.

Objectives

- Clean and validate the customer dataset.
- Explore customer demographics, services, tenure and billing behavior.
- Use SQL and Python to extract and summarize business-relevant information.
- Create visualizations and a Power BI dashboard.
- Apply descriptive and inferential statistics.
- Segment customers using K-Means clustering and PCA.
- Build basic churn prediction models and evaluate them using classification metrics.
- Translate analytical findings into practical retention recommendations.

3. Data Source & Methodology

The project uses the IBM Telco Customer Churn dataset supplied in the project workspace. The cleaned dataset contains 7,043 rows and 21 columns covering customer demographics, services, contract information, tenure, monthly charges, total charges and churn status.

Data preparation

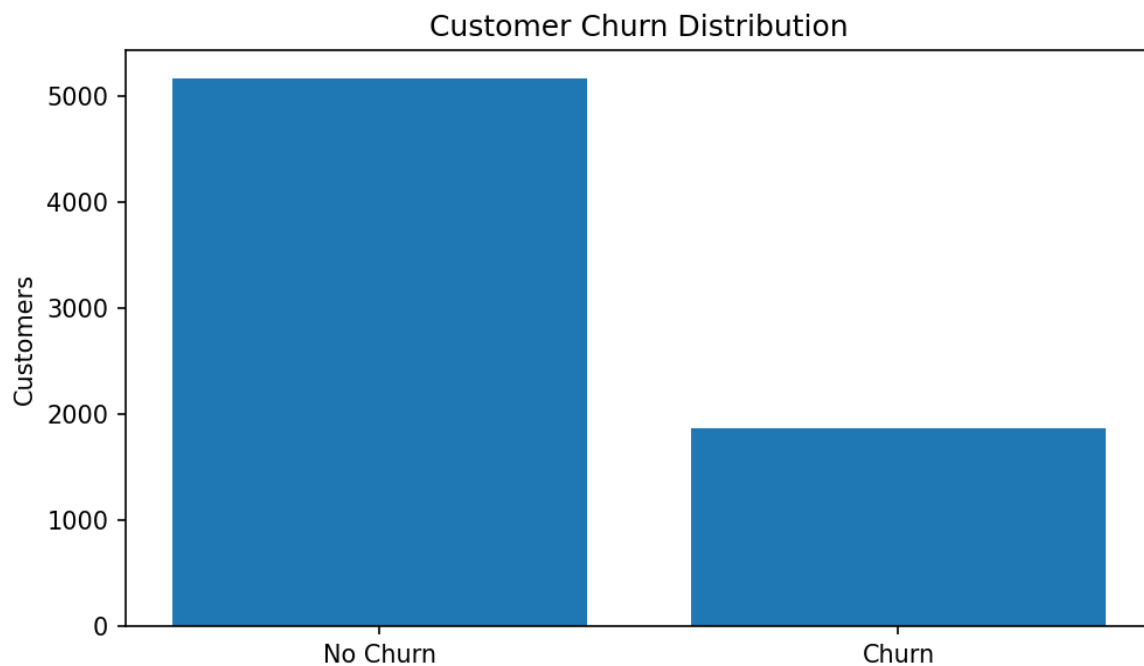
- Standardized column names for consistent Python/SQL usage.
- Converted TotalCharges to numeric values.
- Handled missing TotalCharges values during the cleaning stage.
- Reviewed numerical distributions and potential outliers.
- Prepared separate modeling datasets to avoid feeding missing values into machine-learning algorithms.

Analytical workflow

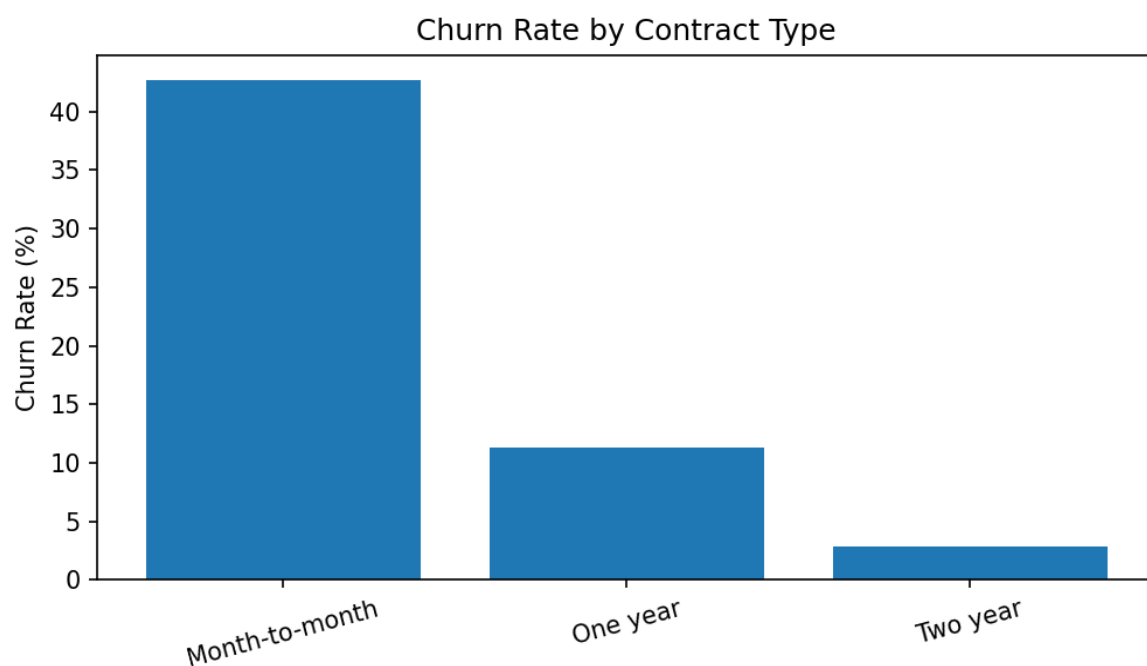
- EDA: distributions, categorical comparisons, correlation analysis and business-oriented visualizations.
- SQL: customer and contract summaries and database-based extraction.
- Visualization: Matplotlib/Seaborn/Plotly and Power BI dashboard development.
- Statistics: descriptive statistics, confidence intervals, t-test, chi-square test, ANOVA and correlation analysis.
- Segmentation: StandardScaler → K-Means → silhouette analysis → PCA visualization → cluster profiling.
- Prediction: train/test split → scaling → Logistic Regression and Decision Tree → classification metrics → feature importance.

4. Key Findings & Exploratory Analysis

The dataset contains 5,174 retained customers and 1,869 churned customers. Overall churn is 26.5%. Average tenure is 32.4 months and average monthly charges are 64.76.



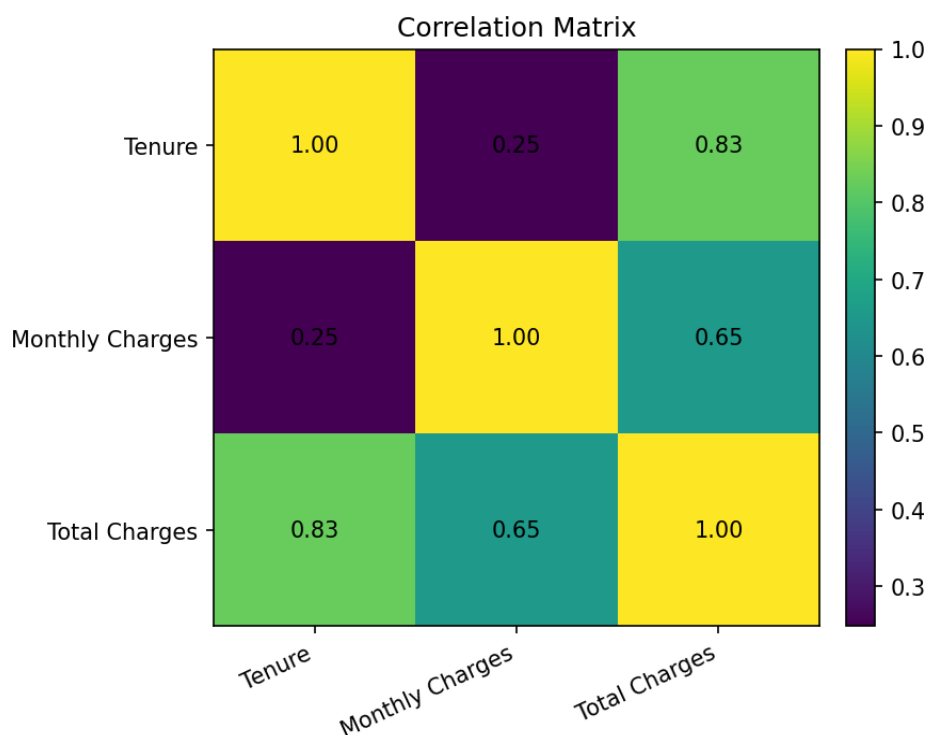
Finding 1: Churn is a substantial minority outcome (26.5%), making retention analysis commercially relevant.



Finding 2: Month-to-month customers show substantially higher churn than customers on one- or two-year contracts.

5. Statistical Analysis

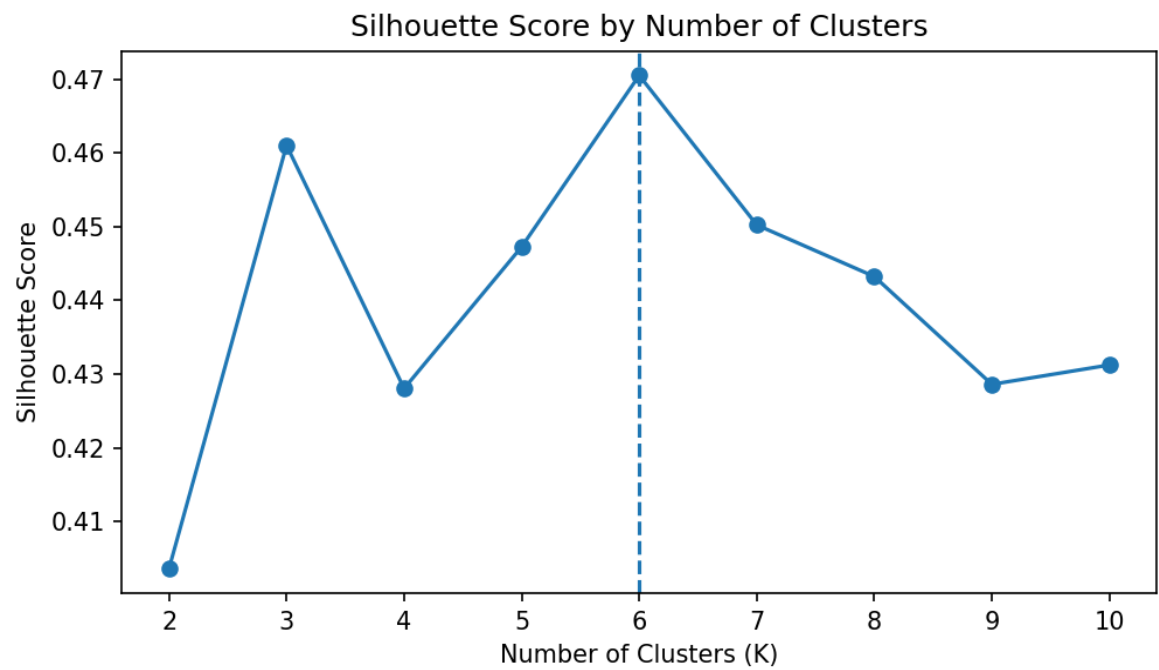
Measure	Result	Interpretation
Average monthly charges	64.76	Central tendency of monthly billing
95% CI for monthly charges	64.06 – 65.46	Likely range for the population mean
Welch t-test: charges vs churn	$t=18.41$, $p<0.001$	Monthly charges differ significantly by churn status
Chi-square: contract vs churn	$\chi^2=1184.60$, $p<0.001$	Contract type is associated with churn
ANOVA: charges by contract	$F=20.83$, $p<0.001$	Mean monthly charges differ across contract groups



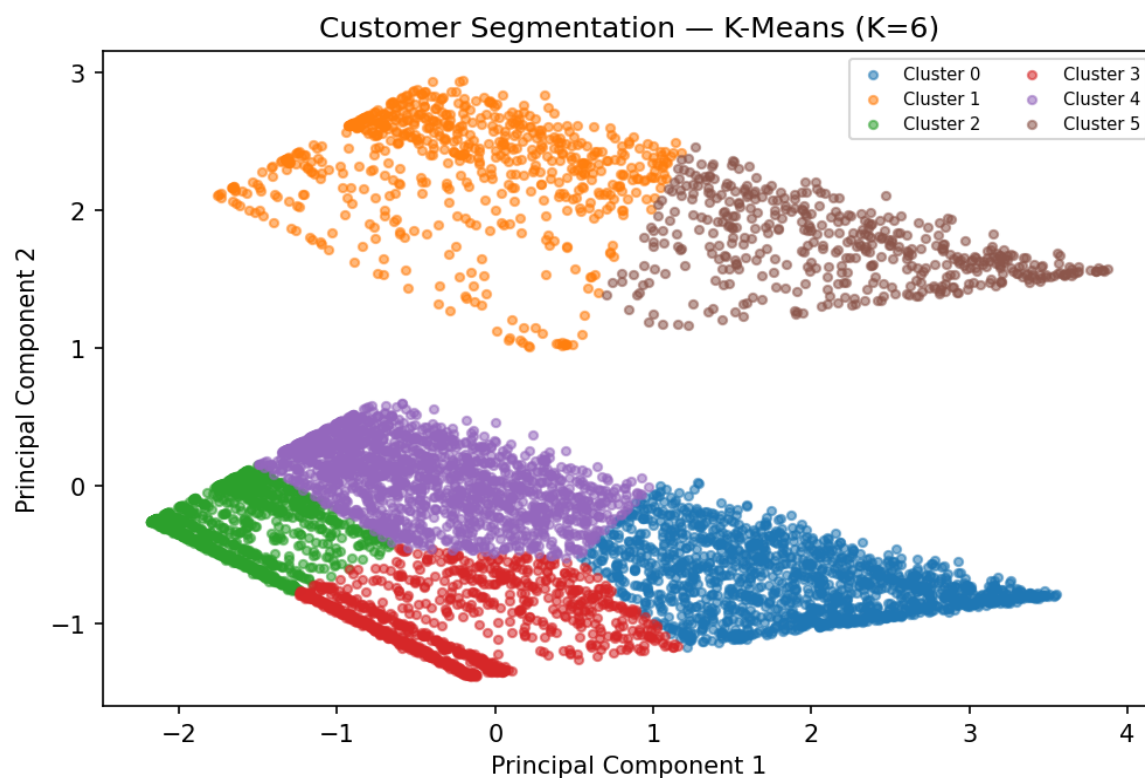
Tenure and total charges show a strong positive correlation ($r=0.826$), which is intuitive because customers who stay longer accumulate more billing. Monthly charges also correlate positively with total charges ($r=0.651$). Correlation should not be interpreted as proof of causation.

6. Customer Segmentation

K-Means clustering was applied to SeniorCitizen, Tenure, MonthlyCharges and TotalCharges after standardization. Silhouette analysis across K=2–10 selected **K=6** as the strongest solution, with a score of 0.471.



Cluster	Customers	Tenure	Monthly Charges	Total Charges
0	1,517	59.9	92.25	5515.95
1	680	17.3	70.71	1118.96
2	1,542	10.1	31.04	291.85
3	1,046	53.4	33.43	1755.31
4	1,785	15.5	79.40	1228.48
5	462	56.9	93.23	5300.13

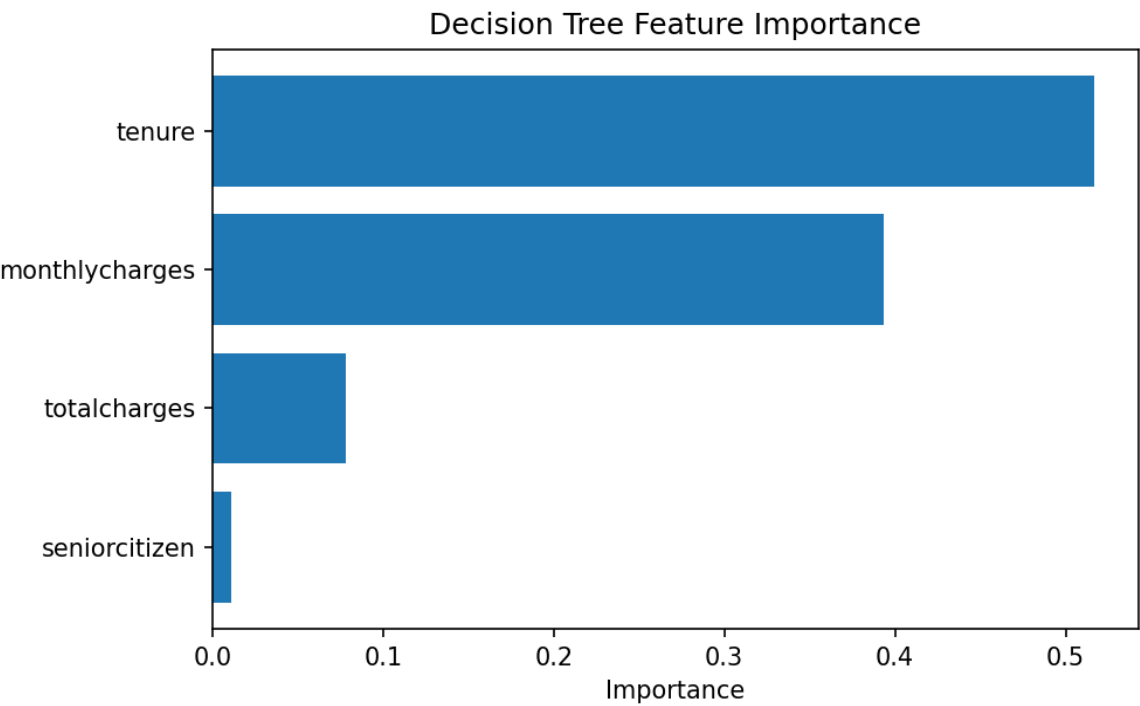


The clusters should be interpreted as behavioral/value groups rather than fixed business labels. For example, groups with long tenure and high charges represent higher-value long-term customers, while low-tenure and low-charge groups represent newer or lower-value customers. The recommended action is to tailor retention and engagement strategies by segment.

7. Predictive Modeling

Churn was encoded as a binary target (1=Yes, 0=No). A stratified 80/20 train-test split was used. The basic models use four numerical features: SeniorCitizen, Tenure, MonthlyCharges and TotalCharges.

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression	0.779	0.616	0.447	0.518	0.816
Decision Tree	0.778	0.609	0.455	0.521	0.807



The Decision Tree indicates that **tenure** (0.517) and **monthly charges** (0.394) are the dominant features in this basic model. This supports the statistical findings that customer tenure and billing level are important signals for churn analysis.

8. Power BI Dashboard

A Power BI dashboard was developed as the interactive reporting layer for the project. The PBIX deliverable contains multiple report pages designed for business exploration rather than only static analysis.

Dashboard Page	Purpose
Summary	Executive view of customer and churn KPIs with comparative visuals.
Customer Details	Customer-level exploration with slicers and key customer attributes.
Churn Reasons	Focused analysis of churn patterns and risk-oriented visualizations.

The final submission should include the PBIX file alongside this PDF report. If a live Power BI Service link is required by the internship portal, the published report URL should be added to the final README/submission form.

Dashboard deliverable filename used in the workspace: **Customer_Churn_Dashboard.pbix**.

9. Recommendations, Limitations & Future Scope

Business Recommendations

- Prioritize retention campaigns for month-to-month customers, who have the highest observed churn rate.
- Monitor customers with high monthly charges and provide targeted plan reviews or value-added offers.
- Create onboarding and early-engagement programs for new customers with short tenure.
- Use customer segments to personalize communication instead of applying a single retention strategy to everyone.
- Use churn scores as decision support rather than automatic decisions, with human review for high-impact actions.

Limitations

- The predictive models use only a limited numerical feature set and are intentionally basic.
- Categorical service and contract variables were not included in the basic predictive model.
- The time-series portion of the internship specification was not applied because the Telco dataset does not contain a genuine date/time variable.
- A stronger production model would require richer feature engineering, validation, calibration and monitoring.
- Historical relationships may change as customer behavior, pricing and service offerings change.

Future Scope

- Add categorical features using one-hot or target-aware encoding.
- Compare additional models such as Random Forest, Gradient Boosting and calibrated classifiers.
- Tune hyperparameters using cross-validation and optimize for recall/PR-AUC when churn detection is the priority.
- Build automated model monitoring and periodic retraining.
- Connect the dashboard to a refreshed data source for near-real-time retention monitoring.

10. Conclusion & Final Deliverables

This project demonstrates an end-to-end data analytics workflow on a customer churn problem: data preparation, exploration, SQL analysis, visualization, statistical inference, segmentation and predictive modeling. The analysis identifies contract type, tenure and billing behavior as important areas for retention strategy.

The final project package consists of the cleaned dataset, Jupyter notebooks, SQL scripts, reports/analysis outputs, Power BI dashboard, and this executive report. Together, these artifacts provide both technical evidence of the analytical process and business-facing insights.

Deliverable	Status
Cleaned Telco dataset	Completed
EDA & visualization notebooks	Completed
SQL analysis scripts	Completed
Statistical analysis	Completed
K-Means segmentation + PCA	Completed
Churn prediction models	Completed
Power BI dashboard (.pbix)	Completed
Final PDF report	Completed

Final takeaway: retention efforts should focus on customers with higher churn-risk signals, especially month-to-month contracts, short tenure and higher monthly charges, while using customer segmentation to personalize the response.